Evaluation runs: **fencing** (fencing-grove)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

No run of the stateless engine on this benchmark.

stateful

No run of the stateful engine on this benchmark.

offline

offline: pair **fenced**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	164	conforms	0.4	8	10 min, 16 GB	0.0	0.4	0.0	in-memory	brain05, 2026-11-02
2	6	660	conforms	0.4	8	10 min, 16 GB	0.1	0.4	0.0	in-memory	brain05, 2026-11-02
3	20	2540	conforms	1.0	8	10 min, 16 GB	0.0	1.0	0.0	in-memory	brain05, 2026-11-02
4	70	9460	conforms	3.3	10	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain05, 2026-11-02
5	252	35004	conforms	9.7	22	10 min, 16 GB	0.1	9.7	0.1	in-memory	brain05, 2026-11-02
6	924	130092	conforms	27.6	51	10 min, 16 GB	0.2	27.2	0.4	in-memory	brain05, 2026-11-02
7	3432	486972	conforms	60.2	180	10 min, 16 GB	0.3	56.0	4.2	in-memory	brain05, 2026-11-02
8	12870	1835700	conforms	233.6	735	10 min, 16 GB	0.8	129.2	104.4	in-memory	brain05, 2026-11-02
9	48620	6961916	🕓 (1 h)			1 h, 250 GB	1.7	588.8		in-memory	brain05, 2026-11-03

offline: pair **stale-frontend**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	176	3	violation	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
2	6	1152	40	violation	0.3	8	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain05, 2026-11-02
3	20	12128	318	violation	2.4	10	10 min, 16 GB	0.0	2.4	0.0	in-memory	brain05, 2026-11-02
4	70	158080	2532	violation	8.4	22	10 min, 16 GB	0.1	8.3	0.1	in-memory	brain05, 2026-11-02
5	252	2230176	20578	violation	68.7	147	10 min, 16 GB	0.1	67.5	1.2	in-memory	brain05, 2026-11-02
6	924	32497920		violation	1196.1	1305	10 min, 16 GB	0.2	1169.0	27.1	in-memory	brain05, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): 6 stale-frontend fifo, 7 fenced fifo, 7 stale-frontend fifo, 11 fenced fifo.

skipped: a smaller size had reached its limit: 7 stale-frontend fifo, 8 stale-frontend fifo, 9 stale-frontend fifo.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **fenced**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	164	conforms	5.3	194	10 min, 16 GB	0.0	5.3	0.0	in-memory	brain01, 2026-11-02
2	6	660	conforms	11.7	6316	10 min, 16 GB	0.1	11.7	0.0	in-memory	brain01, 2026-11-02
3	20	2540	conforms	40.8	8064	10 min, 16 GB	0.0	40.8	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **stale-frontend**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	176	3	violation	4.5	203	10 min, 16 GB	0.0	4.5	0.0	in-memory	brain01, 2026-11-02
2	6	1152	40	violation	11.7	5863	10 min, 16 GB	0.1	11.7	0.0	in-memory	brain01, 2026-11-02
3	20	12128	318	violation	125.3	8170	10 min, 16 GB	0.0	125.3	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Java implementation against the Rust specification). planned but never scheduled (grid points beyond the ladder): 4 fifo fenced, 4 fifo stale-frontend.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **fenced**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	164	conforms	3.6	166	10 min, 16 GB	3.6	0.4	0.0	in-memory	brain01, 2026-11-02
2	6	660	conforms	3.8	166	10 min, 16 GB	3.8	0.4	0.0	in-memory	brain01, 2026-11-02
3	20	2540	conforms	3.8	161	10 min, 16 GB	3.8	1.0	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **stale-frontend**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	176	3	violation	3.6	166	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02
2	6	1152	40	violation	3.8	166	10 min, 16 GB	3.8	0.3	0.0	in-memory	brain01, 2026-11-02
3	20	12128	318	violation	3.8	163	10 min, 16 GB	3.8	2.4	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 4 fifo fenced, 4 fifo stale-frontend.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **fenced**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	164	conforms	3.6	162	1 h, 250 GB		brain01, 2026-11-02
2	6	660	conforms	4.0	162	1 h, 250 GB		brain01, 2026-11-02
3	20	2540	conforms	4.8	157	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **stale-frontend**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	2	176	60	violation	3.6	162	1 h, 250 GB	2	brain01, 2026-11-02
2	6	1152	828	violation	4.1	162	1 h, 250 GB	6	brain01, 2026-11-02
3	20	12128	11076	violation	6.2	157	1 h, 250 GB	19	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **fenced**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	164	164	conforms	5.3	194	10 min, 16 GB	5.3	0.4	0.0	in-memory	brain01, 2026-11-02
2	660	660	conforms	11.8	6316	10 min, 16 GB	11.7	0.4	0.0	in-memory	brain01, 2026-11-02
3	2540	2540	conforms	40.9	8064	10 min, 16 GB	40.8	1.0	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **stale-frontend**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	176	176	conforms	4.5	203	10 min, 16 GB	4.5	0.1	0.0	in-memory	brain01, 2026-11-02
2	1152	1152	conforms	11.7	5863	10 min, 16 GB	11.7	0.3	0.0	in-memory	brain01, 2026-11-02
3	12128	12128	conforms	125.4	8172	10 min, 16 GB	125.3	2.4	0.0	in-memory	brain01, 2026-11-02

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 4 fifo fenced, 4 fifo stale-frontend.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **fenced**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	164	164	conforms	5.3	194	10 min, 16 GB	0.4	5.3	0.0	in-memory	brain01, 2026-11-02
2	660	660	conforms	11.7	6316	10 min, 16 GB	0.4	11.7	0.0	in-memory	brain01, 2026-11-02
3	2540	2540	conforms	40.8	8064	10 min, 16 GB	1.0	40.8	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **stale-frontend**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	176	176	conforms	4.5	203	10 min, 16 GB	0.1	4.5	0.0	in-memory	brain01, 2026-11-02
2	1152	1152	conforms	11.7	5863	10 min, 16 GB	0.3	11.7	0.0	in-memory	brain01, 2026-11-02
3	12128	12128	conforms	125.3	8172	10 min, 16 GB	2.4	125.3	0.0	in-memory	brain01, 2026-11-02

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 4 fifo fenced, 4 fifo stale-frontend.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **fenced**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	164	164	conforms	5.0	239	10 min, 16 GB	5.0	0.1	0.0	in-memory	brain02, 2026-11-03
2	660	660	conforms	10.1	949	10 min, 16 GB	10.1	0.3	0.0	in-memory	brain02, 2026-11-03
3	2540	2540	conforms	38.1	7634	10 min, 16 GB	38.1	1.2	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **stale-frontend**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	176	176	conforms	4.6	204	10 min, 16 GB	4.6	0.1	0.0	in-memory	brain02, 2026-11-03
2	1152	1152	conforms	12.1	5870	10 min, 16 GB	11.9	0.4	0.2	in-memory	brain02, 2026-11-03
3	12128	12128	conforms	109.2	7833	10 min, 16 GB	109.2	2.5	0.0	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): 4 fifo fenced, 4 fifo stale-frontend.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **fenced**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	164	164	conforms	5.0	239	10 min, 16 GB	0.1	5.0	0.0	in-memory	brain02, 2026-11-03
2	660	660	conforms	10.1	949	10 min, 16 GB	0.3	10.1	0.0	in-memory	brain02, 2026-11-03
3	2540	2540	conforms	38.1	7634	10 min, 16 GB	1.2	38.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **stale-frontend**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	176	176	conforms	4.6	204	10 min, 16 GB	0.1	4.6	0.0	in-memory	brain02, 2026-11-03
2	1152	1152	conforms	12.1	5870	10 min, 16 GB	0.4	11.9	0.1	in-memory	brain02, 2026-11-03
3	12128	12128	conforms	109.2	7833	10 min, 16 GB	2.5	109.2	0.0	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). planned but never scheduled (grid points beyond the ladder): 4 fifo fenced, 4 fifo stale-frontend.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **fenced**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	164	164	conforms	5.4	170	1 h, 250 GB		brain02, 2026-11-03
2	660	660	conforms	10.6	942	1 h, 250 GB		brain02, 2026-11-03
3	2540	2540	conforms	42.7	8067	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair stale-frontend, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	176	176	conforms	4.7	211	1 h, 250 GB		brain02, 2026-11-03
2	1152	1152	conforms	13.0	5610	1 h, 250 GB		brain02, 2026-11-03
3	12128	12128	conforms	129.3	8143	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair fenced, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	164	164	conforms	5.7	235	1 h, 250 GB		brain02, 2026-11-03
2	660	660	conforms	10.4	945	1 h, 250 GB		brain02, 2026-11-03
3	2540	2540	conforms	39.2	7630	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair stale-frontend, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	176	176	conforms	4.6	200	1 h, 250 GB		brain02, 2026-11-03
2	1152	1152	conforms	12.3	5866	1 h, 250 GB		brain02, 2026-11-03
3	12128	12128	conforms	111.8	7829	1 h, 250 GB		brain02, 2026-11-03